

Katherina Dayeon

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EDUCATION

University of Central Florida, Burnett Honors College

Bachelor of Science in Computer Science, GPA: 3.8

Orlando, FL

Expected May 2028

EXPERIENCE

Software Engineer Intern

Jan. 2025 – May 2025

Bank of New York Mellon

Orlando, FL

- Developed an anomaly detection app, monitoring service performance metrics like latency using a **ML model** with **real-time** alerts.
- Deployed an automated remediation system resolving **50+** detected anomalies without the use of manual intervention.
- Crafted a frontend dashboard using **React**, translating complex monitoring data into a clean, user-facing interface.
- Streamlined notifications by building an email automation pipeline using **Power Automate** for anomaly alerts.

Software Engineer Intern

Dec. 2025

Institute for Simulation and Training - UCF

Orlando, FL

- Contributed to a **U.S. Department of Energy** & ORTECC research initiative, engineering a smart factory monitoring solution.
- Constructed an interactive 3D factory environment using **Three.js** and simulated sensor data to visualize machines and conditions.
- Embedded **OpenAI**-powered assistance with live machine data to help users interpret system behavior and operational context.
- Boosted frontend response time by **30%** by optimizing data flow between the UI and 3D environment, enhancing usability.

Website & App Developer

Aug. 2025 – Dec. 2025

UCF Arboretum

Orlando, FL

- Collaborated within a team of **5** to plan, develop, and implement internal applications to streamline operations.
- Amplified website traffic by **20%** by redesigning key pages, improving navigation flow, and resolving critical usability issues.
- Produced user-focused interface designs using **Figma** and **HTML**, improving layout structure and overall visual appeal.
- Applied **accessibility** best practices into web design, ensuring an inclusive and improved experience for all site visitors.

Design Director

Jan. 2025 – Dec. 2025

Knight Hacks

Orlando, FL

- Led a design team of **20+** using **Notion** to delegate and track project tasks across all hackathon design needs.
- Refined **10–15** design assets per month using **Adobe Illustrator**, Photoshop, Procreate, and Canva.
- Revamped website and social media presence, reaching **2000+** Instagram followers and increasing follower count by **45%**.
- Rebranded club merch and websites serving **6000+ members**, boosting member outreach and engagement by **50%**.

PROJECTS

LockIn-Buddy | *Tauri, React, FastAPI, Python, OpenCV, Mediapipe, Ollama, faster-whisper, TypeScript, Arduino*

Apr. 2026

- **1st Place Winner** out of **32** projects for the **Checkpoint Award** at **Knight Hacks 2026 Project Launch Program**.
- Collaborated on a cross-platform focus app with a **3D printed** BMO, supporting **17+** REST and WebSocket endpoints in **Python**.
- Designed and crafted a cohesive multi-page UI using **React**, implementing interactive timers and real-time state management.

SMARTLook | *OpenAI, Three.js, REST APIs, React, JavaScript*

Dec. 2025

- Architected a web-based maintenance tool using **React** and **Three.js** to visualize real-time machine data in an interactive 3D space.
- Integrated **OpenAI** via REST APIs to provide AI-powered assistance, system explanations, and contextual user guidance.
- Accelerated 3D rendering from **10s** to under **4s** by applying DRACO mesh compression and reducing frontend data transfer.

Potara | *Gemini, REST APIs, Express.js, MinIO, PostgreSQL, Prisma, Docker, React, JavaScript*

Sept. 2025

- Launched a collaborative art tool powered by **Google Gemini**, providing live AI feedback on a fully interactive web-based canvas.
- Spearheaded the frontend using React, **TailwindCSS**, and Figma, producing a polished and responsive user interface.
- Slashed live AI feedback latency from **15s** to under **5s** by optimizing the data transfer pipeline using base64-encoded image data.

ShapeShift | *React, JavaScript, TypeScript, Figma*

Apr. 2025

- Competed in **Bitcamp 2025**, shipping a full stack web app that converts 2D SVG assets into production-ready 3D models.
- Designed and implemented the **React**-based frontend for a dev tool that streamlines the SVG-to-3D model conversion workflow.
- Prototyped a responsive UI/UX using **React** and **Figma**, increasing overall user engagement by **40%**.

AiSight | *Python, TypeScript, React Native, Figma*

Jan. 2025 – Apr. 2025

- Placed **4th** out of **15** teams in Knight Hacks Project Launch, building accessibility glasses and an app for low-visibility users.
- Engineered a **React Native** mobile app to assist the visually impaired, enhancing accessibility through AI-powered voice commands.
- Partnered with a cross-functional team of **6**, coordinating frontend and backend development for efficient task distribution.

TECHNICAL SKILLS

Languages: HTML, CSS, Python, C, JavaScript, Java, TypeScript, C#

Frameworks: React, React Native, Tailwind, Three.js, Tauri, Express.js, FastAPI, Flutter, TKinter, Unity

Databases & DevOps: PostgreSQL, Prisma, Docker, MinIO

Developer Tools: Git, GitHub, VS Code, Visual Studio, Figma, Blender, Canva, Adobe Illustrator, Power Automate

Libraries & AI: OpenCV, Mediapipe, Ollama, OpenAI, Gemini